Analysis output

My data analysis contained the followimng stages

-The following is the output of my analysis of the data:
step 1 describes how data was pre-processed (raw data cleaning and restructuring)
-Step 2 describes outlier detection (identifying the points that vary widely from the other observations)
-Step 3 describes how missing values were handled (addressing gaps or incomplete parts of the dataset).]
-Step 4 describes how I will use hypothesis testing (testing assumptions and conclusions related to the data).
-Step 5 describes how data visualization will help me to visually display the results of the analysis (charts or graphs showing the insights obtained from the analysis).

	statistic	df	p.value	missing.patterns
1	1809.34	1102	0	54

Figure 1 Little MCAR Test

Figure 1 is a Little MCAR test showing there is a significant statistical result. The calculated test statistic is df(1102) = df (1102), and the p-value = 0; this p-value is far below the common threshold for "significant" p-values (i.e 0.05). Therefore, we rejected the null hypothesis (H_0) and concluded that the pattern of missing values in the dataset is; therefore missing; values are not MCAR. The missingness in the dataset will have an association, i.e., contain evidence showing it can be related both observed or unobserved variables that are either MAR or MNAR in nature. Hence, we conclude that the required approach for effectively managing the

missing values will require a more advanced imputation strategy called multiple imputation.

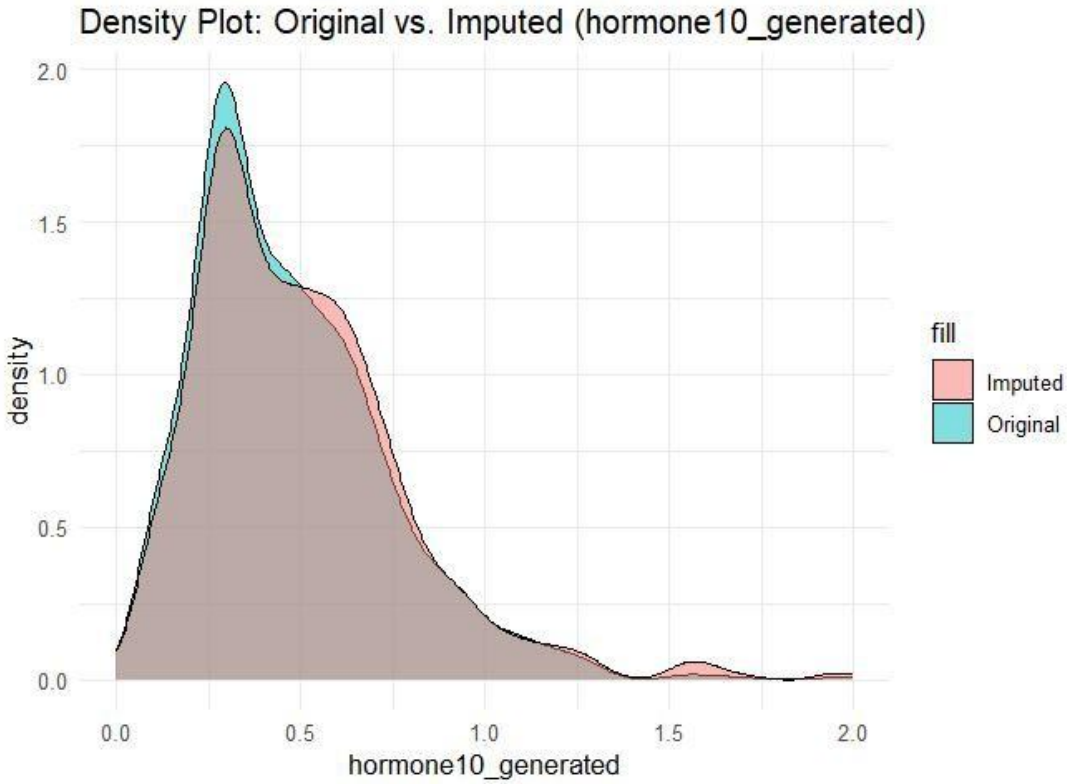

Figure 2 : Hormone Density Plot

Figure 2 is a hormone density plot. The density plot depicts the distribution of the variable called "hormone10_generated" before and after management of the missing values. The blue area depicts the initial variable distribution, whereas the brown area depicts the variable distribution after the completion of the imputation process and all missing values are filled. The final density shape of the before data is very closely aligned with the after data.

We can see that the Density Plot illustrates the distribution of the variable "hormone10_generated" before and after addressing missing values. sThe blue region represents the initial data distribution, while the brown region depicts the distribution following imputation (filling in missing values). The near-identical shapes of these distributions indicate that the imputation method effectively preserved the original data pattern and spread. This suggests a successful imputation process, as the artificial values introduced did not unduly distort the underlying data structure

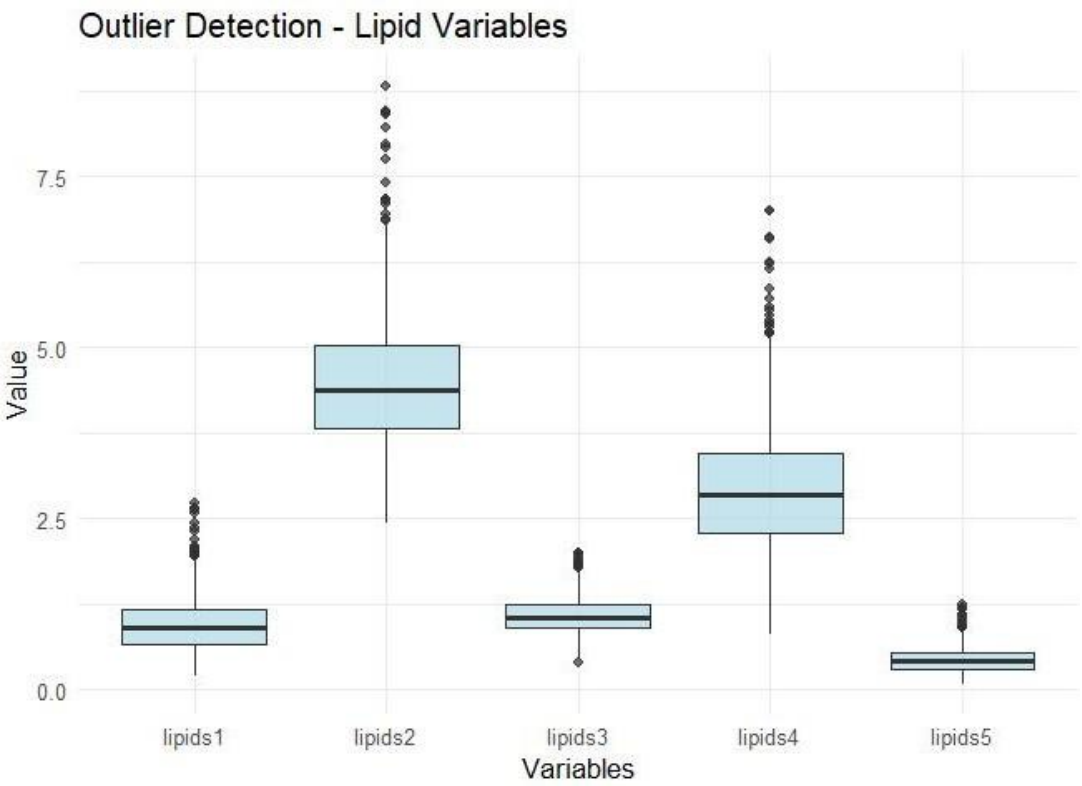

Figure 3 : Outlier Identification boxplot

The figure shows how the five lipids (lipids1-5) differ with regard to the number of outliers (small circles above and below each box). These data points have significant physical differences from the rest of the data and thus indicate they do not fit the general distribution. Outliers were identified in lipids2 and lipids4 by both number and magnitude, providing evidence that lipid2 and

lipid4 have a greater frequency of extreme values than other lipid measurements.

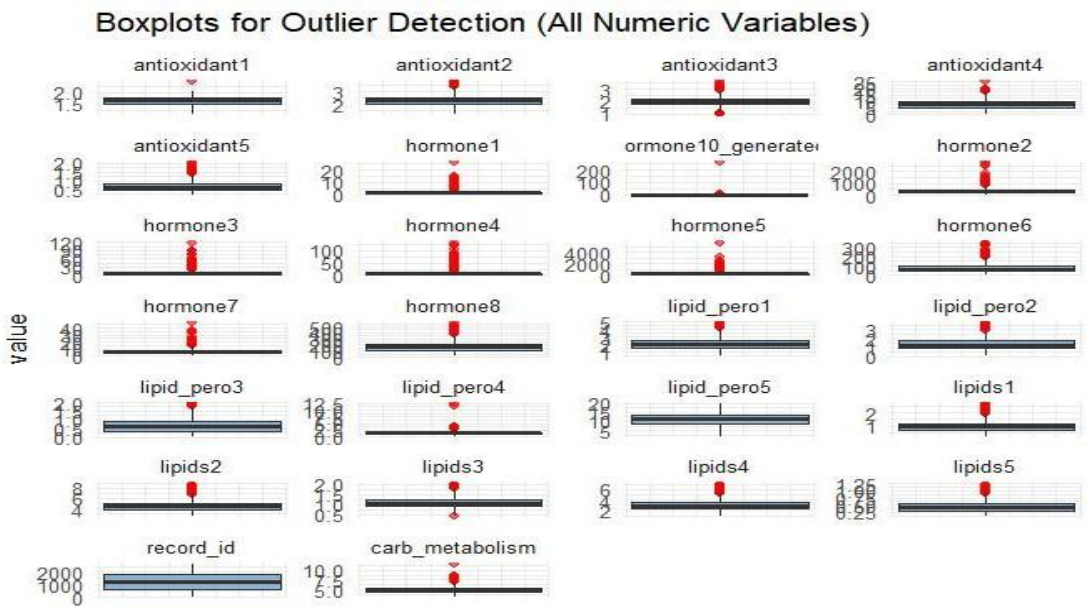

Figure 4 : The Boxplots for Outlier Detection

The figure above reveal that each numerical column in the dataset contains potential outliers. The sBoxplots provide a visual representation of the distribution of data values across several different variables: antioxidant levels (antioxidant1-5), hormone levels (hormone1-8), lipid peroxidation values (lipid_pero1-5), and lipid profiles (lipids1-5). The outliers have been plotted as red dots; these data points have significant physical differences from what would normally be expected for their variable. Outliers were found to be present at high frequency in hormonal and lipid measurements, revealing that there are a large number of extreme data points present in the dataset as a whole.

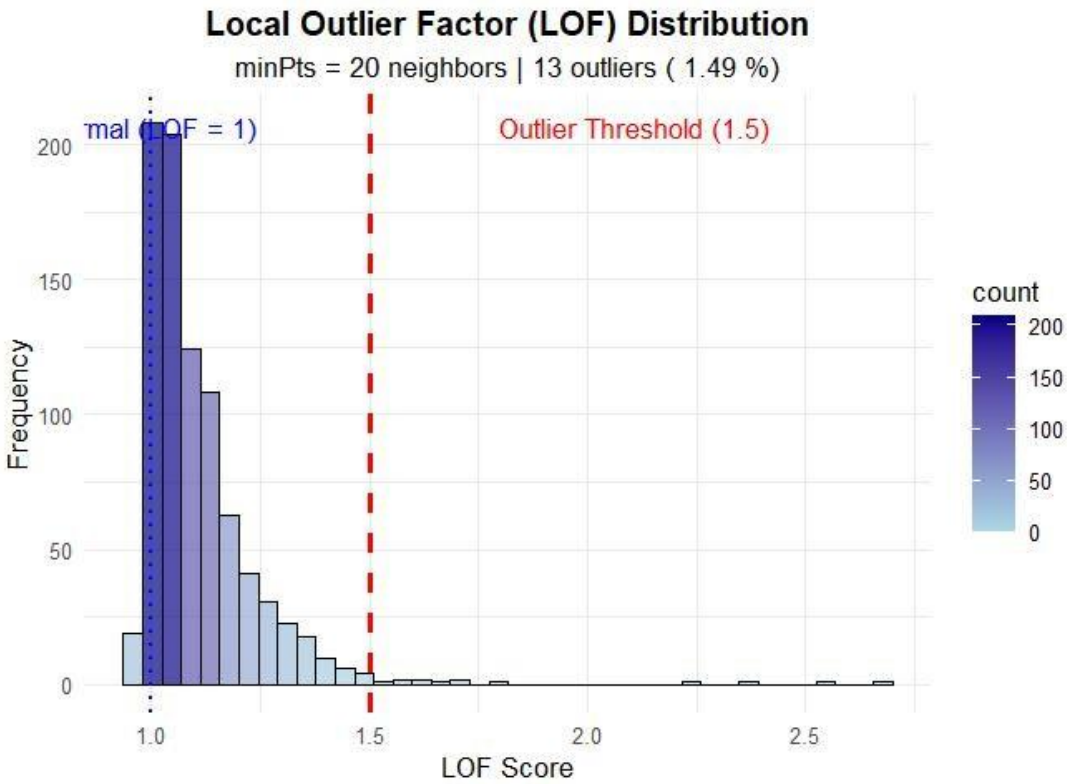

Figure 5 : The Histogram for Outlier Detection

The histogram here display LOF (Local Outlier Factor) scores of all records in the dataset indicate that the vast majority of records cluster around the value 1 (typical value) and have an overall normal distribution. The red dashed line is drawn at the value of 1.5 (outlier threshold). All points in the histogram with values greater than this threshold are considered outliers, in this case there were a total of 13 outliers using 20 neighbors in the LOF calculation which represents 1.49% of the total dataset (indicating that virtually all records in the dataset are consistent and similar to other records; only a very small number of records represent extreme or anomalous values).

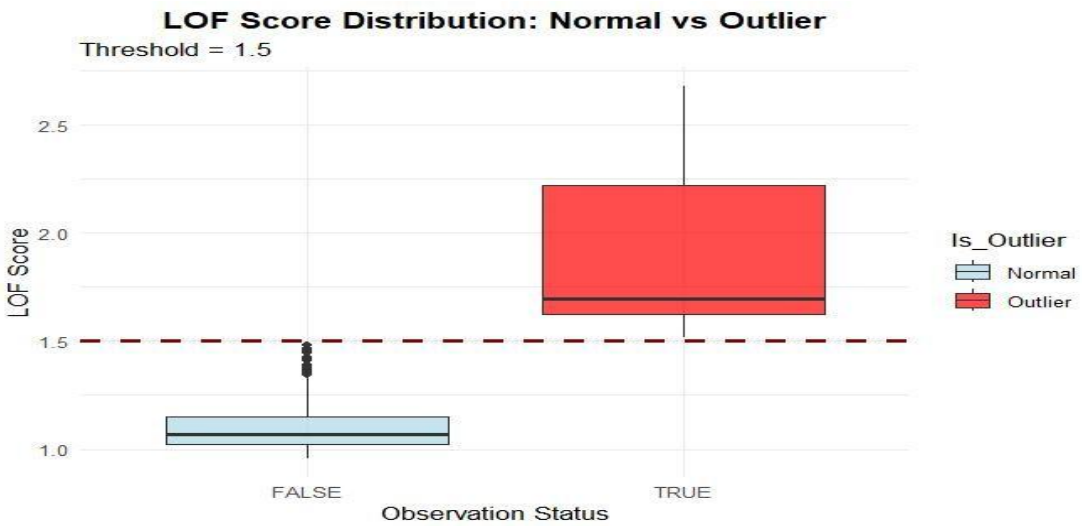

Figure 6 : Boxplots comparing LOF scores

The box plot compares the Local Outlier Factor (LOF) scores between normal data points (to the left) and outlier data points in the righ tand Normal data points shown as false, have a LOF score consistently below 1.5, represented by a red dashed line, meaning they fit the overall pattern of the data and don't have any unusual characteristics. On the other hand, the LOF scores for outlier data points shown as true, are consistently greater than 1.5, indicating that these data points are significantly different from the normal sample of data points. The definition of separation between the two groups of points is very clear and supports the conclusion that the method used to detect outliers in this dataset is effective.

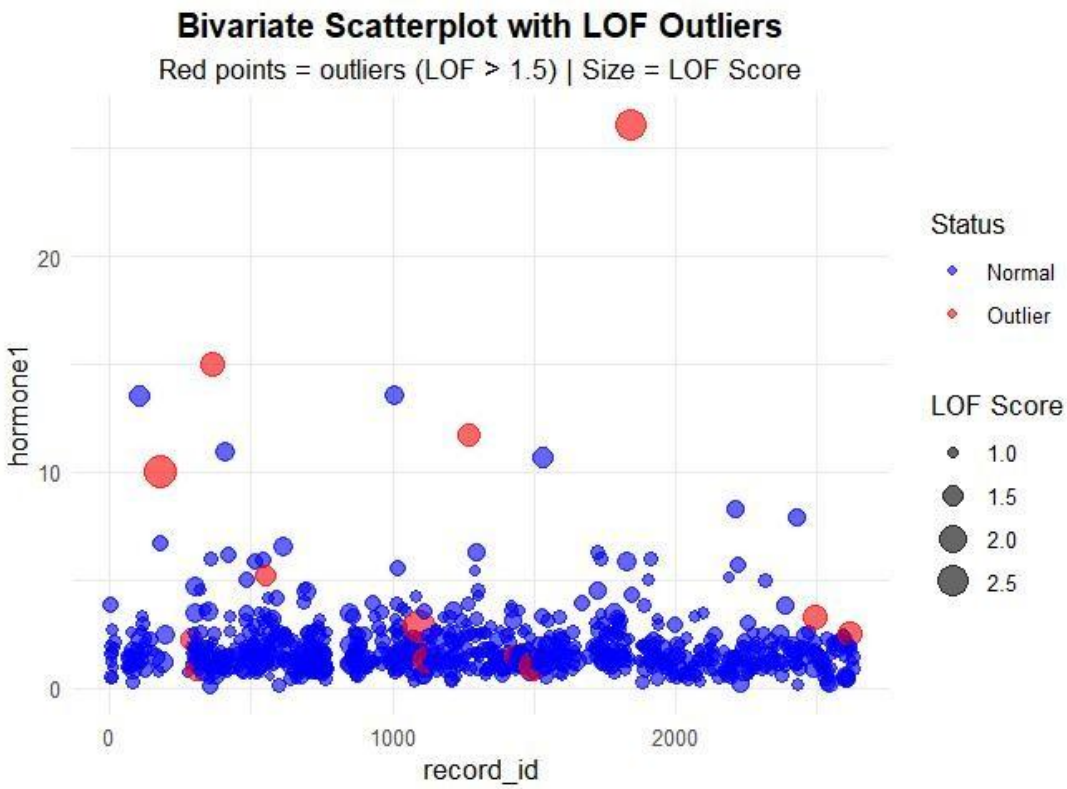

Figure 7: Bivariate Scatterolot

Here the scatterplot of hormone1 level across the unique record identifiers shows blue points where normal observations exist, while red points indicate outliers as determined by the LOF algorithm with LOF score of > 1.5. The major trend of the data points is found in a dense cluster towards the lower end of the hormone1 scale (values between approximately 0 and 5). However, there are several red points that are very high, indicating there are outlier spikes in hormone levels. The darker the color, the greater the amount of deviation from the norm, and the size of the dot indicates strength of the outlier status of the point.

Conclusion

The missing data mechanism appears to be consistent with either Missing at Random (MAR) or Missing Not at Random (MNAR), but not Missing Completely at Random (MCAR). The PNPM imputation has been done accurately to a high degree of accuracy, multivariate outlier points have been identified, and data were visualized.